

Long-tail loss-side methods for BST: focal loss in depth, plus an idiot's guide for the rest

For ML team. Companion to Tasks 1 (pose-anchor continuity) and 2 (skeleton augmentation).

TL;DR

Focal loss with class weighting. $\gamma=2.0$, α from the existing inverse-frequency class weights as the brief specifies. Flagged for awareness: if Task 2's v-sampler is also live, total rare-class amplification compounds to $\sim 38\times$ (sampler $\sim 3.5\times \times \alpha \sim 11\times$) which is past class-balanced and a plausible overfitting risk on the 358 Rush clips. If that bites in the first run (rare-class train F1 climbs while val F1 doesn't), two fallbacks: revert to uniform sampler for the focal run, or scale α to $1/vcount$ so total recovers to $\sim 11\times$, matching a class-balanced reference. Defaulting to the brief; switch only if the diagnostic shows the compounding biting.

Keep label smoothing ($\epsilon=0.1$) and the rest of the loop as-is for the first focal run. Focal and label smoothing both pull on confident examples; the combined interaction wasn't directly tested in Mukhoti et al. (their comparison is focal vs label-smoothed-CE as alternative losses, not focal+LS combined), so it's genuinely unstudied. Bundling the change destroys ablation interpretability. Flag label smoothing as a follow-up ablation if focal alone underdelivers.

Three other family-3 methods (logit adjustment, balanced softmax, class-balanced loss) covered in part 3 as reference. Logit adjustment is the cleanest next attempt if focal underdelivers. Don't combine multiple loss-side methods at once.

1. Why loss-side at all

Sampling and augmentation change what the model sees per epoch. Loss-side methods change how the gradient updates respond to what the model sees. They're complementary, not substitutes.

- **Augmentation:** expands the effective training signal. Same labels, more variety per class. Doesn't directly target imbalance, that's the `WeightedRandomSampler`'s job, which is sampling-side.
- **Sampling (`WeightedRandomSampler`):** changes the *rate* at which classes appear in minibatches. Rare classes show up more often; the model sees more rare-class gradients per epoch.
- **Loss-side (this memo):** changes the *magnitude* of the gradient update for each example. Two flavours: per-class reweighting (rare-class examples get bigger gradients), and per-example reweighting (hard examples get bigger gradients). Focal loss does both.

What's already in the loop (per `tuning_thoughts.md`): cross-entropy with label smoothing $\epsilon=0.1$ and class-weighted CE. Class-weighted CE is the simplest per-class reweighting; it multiplies the standard CE by a per-class scalar. Label smoothing is a different beast: it caps target confidence at $1-\epsilon$ on the true class, which is a regulariser against overconfidence rather than a long-tail tool. Both stay; focal loss either replaces or composes with class-weighted CE depending on framing (part 2.5).

The reason to add focal loss on top: class-weighted CE treats every example in a class identically. A correctly-classified Rush at 0.95 confidence and a misclassified Rush at 0.10 confidence get the same per-class weight, but the second is where the gradient signal we care about lives. Focal preferentially amplifies the second.

2. Focal loss with class weighting

2.1 What it is

Focal loss reshapes cross-entropy with a focusing term:

$$FL(p_t) = -\alpha_t \cdot (1 - p_t)^\gamma \cdot \log(p_t)$$

where p_t is the model's predicted probability for the true class on that example, $\gamma \geq 0$ is the focusing parameter, and α_t is the per-class weight. The $(1 - p_t)^\gamma$ term is the modulator: when p_t is near 1 (easy example, model confident), $(1 - p_t)^\gamma$ is near 0 and the loss collapses. When p_t is near 0 (hard example, model very wrong), $(1 - p_t)^\gamma$ is near 1 and the loss is roughly cross-entropy. $\gamma=0$ recovers vanilla cross-entropy exactly; $\alpha_t=1$ recovers unweighted focal.

Two knobs: γ is focusing strength (how aggressively to discount easy examples), α_t is per-class weight (how strongly to upweight rare-class examples). They're orthogonal: γ acts per-example based on confidence, α_t acts per-class based on label frequency.

2.2 Intuition and why it helps long-tail

Two mechanisms, both relevant here:

1. **Hard-example focus (γ).** In a long-tailed dataset, tail-class examples are typically *also* the hard examples. The model sees them rarely, so confidence on them stays low for longer in training. Focal's per-example modulation amplifies tail-class gradients without ever needing to know labels are imbalanced. This is the original RetinaNet motivation: dense object detection has overwhelming background-vs-object imbalance, and focal lets easy-background examples discount themselves automatically.
2. **Per-class reweighting (α).** Direct intervention: bigger gradient for tail-class examples regardless of confidence.

The combination matters. With γ alone, a misclassified head-class example also gets gradient amplification; fine, but not specifically targeted at long-tail. With α alone (= class-weighted CE), every rare-class example gets equal weight, including easy ones that don't need extra signal. Together: rare-class hard examples get the largest gradient kick, easy head-class examples get the smallest.

2.3 Parameters for our regime

Parameter	Value	Reasoning
γ	2.0	RetinaNet's published default; the focal loss paper finds $\gamma=2$ robust across $\gamma \in [0.5, 5]$. Start here, ablate $\gamma \in \{1.0, 2.0, 3.0\}$ once first run is in.

Parameter	Value	Reasoning
α_c (default)	$\propto 1 / \text{class_count_c}$	Existing inverse-frequency weights from the training loop. Use as-is.
α_c (fallback, if compounding bites)	$\propto 1 / \sqrt{\text{class_count_c}}$	$\sqrt{v}$ -scale-down to recover $\sim 11\times$ total amplification when Task 2's $\sqrt{v}$ -sampler is also live. Switch only on the part 4 diagnostic. Math in part 2.4.
Label smoothing ϵ	0.1 (unchanged)	Keep for the first focal run; flagged for downstream ablation if focal alone underdelivers (part 2.6 caveat 3).

2.4 Compounding amplification math (and when to switch to $\sqrt{v}$ -scaled α)

The 14-class imbalance is roughly $12\times$ (Rush 358 vs Net shot 4000+). Four relevant amplification regimes for the rare-class effective gradient signal, computed as (sampler amplification) $\times$ (loss-weight amplification), measured relative to a head class:

Regime	Sampler amp	α amp	Total
Baseline (uniform sampler, uniform α)	$1\times$	$1\times$	$1\times$
Class-balanced reference (uniform sampler, $\alpha = 1/\text{count}$)	$1\times$	$\sim 11\times$	$\sim 11\times$
Task 2 sampler + α ($1/\text{count}$)	$\sim 3.5\times$	$\sim 11\times$	$\sim 38\times$
Task 2 sampler + $\sqrt{v}$ -scaled α ($1/\sqrt{\text{count}}$)	$\sim 3.5\times$	$\sim 3.3\times$	$\sim 11\times$

The third row is the compounding problem. With 358 Rush clips, a $38\times$ gradient amplification means each Rush clip contributes the gradient signal that 38 head-class clips would. That's well past "class-balanced" and into "rare-class-dominated" territory; expected failure mode is overfitting to the specific Rush clips in the train set (rare-class train F1 climbs, val F1 doesn't follow).

The fourth row, $\sqrt{v}$ -scaled α composed with $\sqrt{v}$ -scaled sampler, recovers a $11\times$ total amplification, which matches the class-balanced reference. The two $\sqrt{v}$ -factors compose multiplicatively to give the same net rebalancing as one full inverse-frequency factor on its own. The principled compositional choice: total amplification is the product, so each component takes a square-root share.

Sampler interaction note. $\alpha=1/\text{count}$ is the default whether or not Task 2's sampler is live; no re-derivation needed when the sampler ships. The diagnostic in the previous paragraph governs the only switch.

2.5 Composition with what's already in place

Existing component	Interaction with focal loss
Cross-entropy	Replaced. Focal is a reshaped cross-entropy; $\gamma=0$ reduces exactly. No double-counting.
Class-weighted CE (per-class scalars)	Replaced. The α_t in focal is the per-class scalar. Don't apply both. Migration: take the existing class weights and feed in directly as α . If the

Existing component	Interaction with focal loss
	diagnostic in §4 shows compounding biting (rare-class train F1 climbs without val following), switch α to $1/\sqrt{\text{count}}$ or revert the Task 2 sampler (see 2.4)
Label smoothing ($\epsilon=0.1$)	Composes, but with opposing pressure. Label smoothing caps target confidence at $1-\epsilon$; focal down-weights examples where the model is already confident. Both reduce the contribution of well-classified examples through different mechanisms. Keep $\epsilon=0.1$ for the first focal run (changing two things at once destroys ablation interpretability), and flag $\epsilon=0.05$ or $\epsilon=0$ as a clean follow-up ablation if focal underdelivers.

2.6 Implementation sketch

```
import tensorflow as tf

class ClassWeightedFocalLoss(tf.keras.losses.Loss):
    """
    Class-weighted focal loss with optional label smoothing.
    alpha: tensor of shape (num_classes,), per-class weights.
           Pass normalised (e.g. alpha / mean(alpha)) so  $\gamma$ -ablations
           don't also shift effective LR.
    gamma: focusing parameter;  $\gamma=0$  reduces to weighted CE.
    label_smoothing:  $\epsilon$  for the CE term; 0 disables.
    """
    def __init__(self, alpha, gamma=2.0, label_smoothing=0.0, **kwargs):
        super().__init__(**kwargs)
        self.alpha = tf.constant(alpha, dtype=tf.float32)
        self.gamma = gamma
        self.label_smoothing = label_smoothing

    def call(self, y_true, logits):
        y_true = tf.cast(y_true, tf.int32)
        num_classes = tf.shape(logits)[-1]

        # Smoothed CE per example, no reduction
        y_onehot = tf.one_hot(y_true, num_classes)
        if self.label_smoothing > 0:
            y_target = y_onehot * (1.0 - self.label_smoothing) \
                + self.label_smoothing / tf.cast(num_classes, tf.float32)
        else:
            y_target = y_onehot
        ce = tf.nn.softmax_cross_entropy_with_logits(y_target, logits)

        # True-class probability (unsmoothed) for the focal modulator.
        # Computed via log_softmax + gather rather than exp(-ce), because
        # under label smoothing CE  $\neq -\log(p_t)$  and the shortcut is wrong.
        log_probs = tf.nn.log_softmax(logits, axis=-1)
        p_t = tf.exp(tf.gather(log_probs, y_true, batch_dims=1))
        focal_term = tf.pow(1.0 - p_t, self.gamma)

        alpha_t = tf.gather(self.alpha, y_true)
        return tf.reduce_mean(alpha_t * focal_term * ce)
```

First, this implementation multiplies the focal modulator by the smoothed CE. That's one defensible way to combine focal with label smoothing, not a canonical formulation (there isn't one; see part 2.7 caveat 4). Second, computing p_t via `log_softmax + gather` is more verbose than `exp(-ce)` but is the right move

under label smoothing, since the smoothed CE is no longer equal to $-\log(p_t)$ and the shortcut would silently corrupt the modulator. Third, α should be passed as a normalised tensor (e.g. divided by its mean) so swapping γ doesn't also shift the effective learning rate across runs.

2.7 Caveats

1. **γ tuning matters more than the literature suggests.** RetinaNet's $\gamma=2$ default was tuned for the foreground/background object-detection regime (10^4 to $10^5\times$ imbalance, two-class). At our scale (12 $\times$, 14-class), $\gamma=2$ is a starting point not a finishing one. $\gamma \in \{1.0, 2.0, 3.0\}$ is the natural ablation grid. $\gamma=0$ reduces to weighted CE, so focal is *strictly more general* than the current setup; no scenario where focal performs worse than the right α_t with γ tuned correctly.
2. **The calibration story runs the opposite direction to the common claim.** Common claim: focal loss hurts calibration because the modulator suppresses loss on confident-correct examples, so the model never gets pulled to high confidence. Mukhoti et al. ran the experiment carefully and found the opposite. Focal-loss-trained models are better calibrated than cross-entropy on CIFAR-10/100 and Tiny-ImageNet, including against cross-entropy with label smoothing as a standalone alternative. They argue focal acts as an implicit entropy regulariser, which improves calibration. Note that the paper compares focal alone against [CE, label-smoothed CE, MMCE, Brier] as separate baseline losses and it does not directly test focal combined with label smoothing, so that specific interaction is genuinely unstudied rather than empirically settled. Worth measuring ECE alongside F1 in the ablation; downgrade this concern if calibration looks acceptable.
3. **No skeleton-AR-specific evidence for focal loss on long-tail.** The focal-loss-on-long-tail literature is mostly image classification. Closest skeleton-AR work I found applies focal loss in non-long-tail settings. So $\gamma=2.0$ is literature-anchored, not dataset-tuned. Standard caveat: anchor first, ablate later.
4. **Don't combine with another loss-side method in the same run.** If focal underdelivers, the next step is logit adjustment alone (part 3.1), not focal + logit adjustment. Layered loss-side methods are hard to debug.

3. The rest of family 3: idiot's guide

Three more loss-side methods from the long-tail image-classification literature 2019 to 2021. Reference, not action items.

3.1 Logit adjustment

What it is. Add $\log(\text{class_count}_c / \text{total})$ to the logit for class c *before* the softmax during training. Equivalent to subtracting that quantity from the logit at inference. In one line:

```
logits_adjusted = logits +  $\tau$  * log(class_priors)
```

where `class_priors` is the per-class frequency in training data and τ is a scaling hyperparameter (default 1.0). Loss is otherwise standard cross-entropy on the adjusted logits.

Intuition. Naive softmax CE pushes the model's predicted class probability to match the empirical (long-tailed) training distribution. Logit adjustment shifts the decision boundary by exactly the prior-ratio so the trained model is implicitly Bayes-optimal under a *uniform* test distribution, which is what we evaluate against. Clean derivation from Bayes rule.

Catch. Assumes test-time class distribution is uniform. If your val set is also long-tailed (matches train), logit adjustment over-corrects.

Recommendation. Cleanest fallback if focal loss underdelivers. Replaces focal entirely (don't compose). Single hyperparameter (τ). Easy to add: a few lines on top of standard CE.

3.2 Balanced softmax

What it is. Almost identical to logit adjustment in the loss form: shift the logits by $\log(\text{class_count_c})$ before softmax. The original paper derives it from a slightly different starting point (label distribution shift between train and test) and pairs it with a meta-sampler component, but the standalone Balanced Softmax loss is essentially logit adjustment with $\tau=1$.

```
logits = logits + log(samples_per_class)
loss = F.cross_entropy(logits, targets)
```

Intuition. Same as logit adjustment. Correct for train/test distribution mismatch by shifting logits.

Catch. Same as logit adjustment.

Recommendation. If you go logit-adjustment direction, pick one of these two. Logit adjustment has more explicit τ -tuning; Balanced Softmax is the cleaner one-liner. Equivalent in expectation for our purposes.

3.3 Class-balanced loss / effective number of samples

What it is. Replace per-class weight $1/\text{count_c}$ with $(1 - \beta) / (1 - \beta^{\text{count_c}})$, where $\beta \in [0, 1]$ is a hyperparameter (typical 0.99 or 0.999). The fraction is the "effective number of samples": the marginal value of the 1000th example in a class is much smaller than the marginal value of the 10th, so naive $1/\text{count}$ over-amplifies very rare classes. The original paper plugs this in as the α_t in focal loss (CB-Focal).

Intuition. Diminishing returns on per-class samples.

Catch. Yet another hyperparameter (β). On our 12× imbalance with $n_{\text{head}} \approx 4000$ and $n_{\text{tail}} \approx 358$, $\beta=0.999$ produces a rare-class amplification of $\approx 3.3\times$ via $(1-\beta^{n_{\text{head}}})/(1-\beta^{n_{\text{tail}}}) \approx 982/301$. That essentially matches $\sqrt{v}$ -frequency weighting; CB at $\beta=0.999$ isn't doing meaningful work on top of what we'd already get from $\sqrt{v}$ - α . To genuinely sit between $\sqrt{v}$ -frequency ($\sim 3.3\times$) and inverse-frequency ($\sim 11\times$), β needs to climb to ≈ 0.9999 , which gives $\sim 9\times$ amplification at our scale. The method shines more at 100×+ imbalances (iNaturalist) where inverse-frequency is genuinely catastrophic and the $(1-\beta^n)$ saturation buys you something.

Recommendation. Skip. At our scale CB lands in the same neighbourhood as the α weights we'd use anyway ($\beta \approx 0.9999 \approx 1/\text{count}$, $\beta \approx 0.999 \approx 1/\sqrt{\text{count}}$), so it adds a tuning knob without buying anything. Revisit only if amateur data widens the imbalance significantly.

3.4 Forward pointer: two-stage training

Not family 3 (this is a training-procedure intervention, family 4) but it's the natural next step if focal loss doesn't deliver, so flagging here.

What it is. Train feature extractor and classifier in two phases. Phase 1: train end-to-end with instance-balanced sampling (uniform sampler, no rebalancing), best for representation learning. Phase 2: freeze the feature extractor, retrain only the classifier head on a class-balanced sampler, best for the classifier. Two variants (cRT and LWS) for the classifier-retraining step.

Why it's relevant. The Kang et al. core finding is that *features and classifiers want different sampling regimes*. If our long-tail problem is dominated by the classifier learning a biased decision boundary (likely) but the features are basically fine, two-stage training fixes it surgically. Loss-side methods (focal, logit adjustment) try to fix both at once, which can be suboptimal if the issue is really classifier-side.

When to consider. If focal loss + Task 2 sampler lifts rare-class F1 but not enough, and val/test diagnostics suggest "features are good but classifier is biased," two-stage is the cleanest next move. Bigger engineering lift than focal (needs a training-loop refactor) but conceptually the strongest tool in the long-tail toolbox.

4. Recommendation

Action: replace class-weighted CE with class-weighted focal loss ($\gamma=2.0$, α from the existing inverse-frequency weights) on the next training run. Keep label smoothing $\epsilon=0.1$ unchanged. Don't change anything else. Watch the rare-class train-vs-val F1 gap: if train climbs without val following, the compounding with Task 2's v-sampler is biting and α should drop to $1/v_{\text{count}}$ or the sampler should revert to uniform (see part 2.4).

Validation criterion. Per-class F1 on rare classes (Rush, Long service, Wrist smash) should lift relative to the current weighted-CE baseline; macro F1 should also lift. Overall accuracy may dip slightly because focal pulls the decision boundary toward the tail; if accuracy drops by more than $\sim 2\%$ with no compensating macro F1 gain, γ is too aggressive, try $\gamma=1.0$ next. ECE worth measuring as a sanity check on the calibration story (part 2.7 caveat 2) but not gating.

Default $\alpha=1/\text{count}$ works in either case (sampler on or off). The compounding risk only kicks in when both Task 2's v-sampler and $\alpha=1/\text{count}$ are live together, and even then, the fallback is to switch one of the two, not to pre-emptively v-scale.

If focal underdelivers. Two next moves, in order:

1. Drop label smoothing (try $\epsilon=0.05$ then $\epsilon=0$). Removes the opposing-pressure interaction with focal. Single-knob ablation, fast to run.
2. Replace focal with logit adjustment (part 3.1). Different mechanism (margin-based, not modulation-based), different failure mode. If the issue is "focal is underweighting head classes too much" rather than "rare classes need more help," logit adjustment usually flips the right way.

Don't. Don't combine focal + logit adjustment + class-balanced loss + balanced softmax. They overlap functionally and the failure mode of "all four are on, model is worse, can't tell why" is easy to fall into. One loss-side method at a time.

5. Reconciliation with existing pipeline

Source	States	Recommendation	Status
Current training loop	CE with label smoothing $\epsilon=0.1$, class-weighted with per-class scalar weights	Replace class-weighted CE with class-weighted focal loss. Keep label smoothing $\epsilon=0.1$ unchanged.	Drop-in replacement. The α_t in focal is the per-class weight; no double-counting.
Task 2 (skeleton augmentation)	WeightedRandomSampler with weights $\propto 1/\sqrt{\text{count}}$	Default: α = existing inverse-frequency, sampler stays. Flag for monitoring compounding to $\sim 38\times$ total is the failure mode that triggers either $\alpha \rightarrow 1/\sqrt{\text{count}}$ or sampler reversion.	Compositional with risk; diagnostic in part 4 governs the switch. Math in part 2.4.
tuning_thoughts.md	Class-weighted CE + label smoothing already specified	Replace class-weighted CE component, retain label smoothing	Aligns. Label smoothing flagged as follow-up ablation if focal underdelivers.
Task 1 (pose-anchor)	Independent	No interaction	Different layer (data preprocessing vs loss).
Arch doc Phase 3 plans	Amateur generalisation, wider augmentation parameters	Loss-side methods orthogonal to Phase 3 distribution shift work	Phase 3 is data-distribution work; loss-side stays the same. May need to re-derive α weights if the train-set class balance changes meaningfully under Phase 3 data additions.

6. Open items for Ari

1. **Verify val/test class balance.** Logit adjustment / balanced softmax assume uniform test distribution. If our val set is also long-tailed, the assumption breaks. Doesn't affect focal loss directly; flagging in case logit adjustment becomes the fallback in part 4.

2. **α normalisation convention.** I've recommended dividing α by its mean so that swapping γ doesn't change effective LR. Some implementations normalise α to sum to num_classes; some don't normalise at all. Pick a convention before running so γ -ablation results are interpretable across runs.
3. **ECE measurement.** Worth adding to the eval pipeline regardless; the focal calibration question (part 2.7 caveat 2) wants it. ECE is ~15 lines in TF. Bin predicted confidences, compute |accuracy – confidence| per bin, weighted average. No need to pull a dependency for it."

7. Papers worth having on file

- Lin, Goyal, Girshick, He, Dollár. *Focal Loss for Dense Object Detection*. ICCV 2017, arXiv:1708.02002. The original.
- Mukhoti, Kulharia, Sanyal, Golodetz, Torr, Dokania. *Calibrating Deep Neural Networks using Focal Loss*. NeurIPS 2020, arXiv:2002.09437. Reverses the folk-wisdom calibration claim.
- Menon, Jayasumana, Rawat, Jain, Veit, Kumar. *Long-tail learning via logit adjustment*. ICLR 2021, arXiv:2007.07314. The cleanest fallback; principled derivation.
- Ren, Yu, Sheng, Ma, Zhao, Yi, Li. *Balanced Meta-Softmax for Long-Tailed Visual Recognition*. NeurIPS 2020, arXiv:2007.10740. Sister to logit adjustment.
- Cui, Jia, Lin, Song, Belongie. *Class-Balanced Loss Based on Effective Number of Samples*. CVPR 2019, arXiv:1901.05555. Effective-number reweighting; CB-Focal variant.
- Kang, Xie, Rohrbach, Yan, Gordo, Feng, Kalantidis. *Decoupling Representation and Classifier for Long-Tailed Recognition*. ICLR 2020, arXiv:1910.09217. Two-stage; family-4 forward pointer.